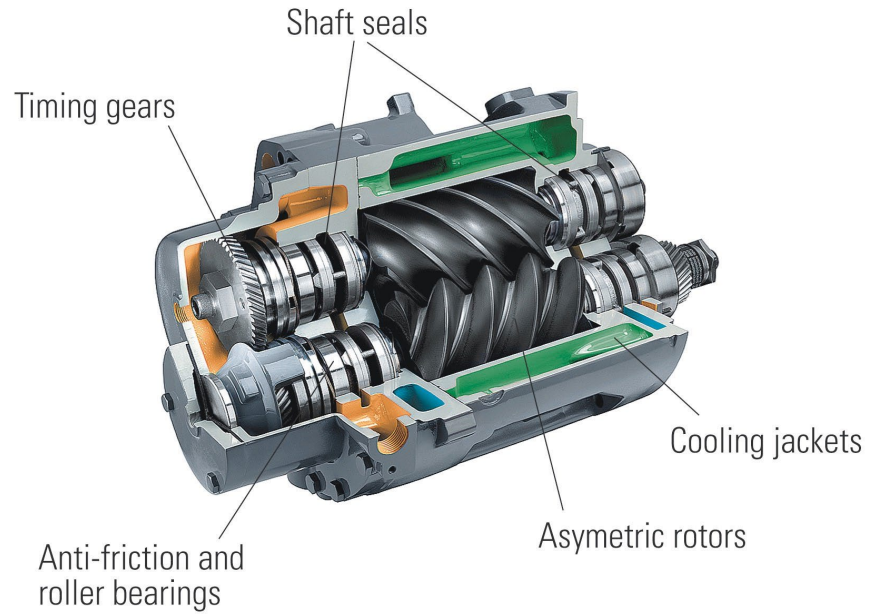




ME 300:
Engineering Thermodynamics

Compressors and Turbines

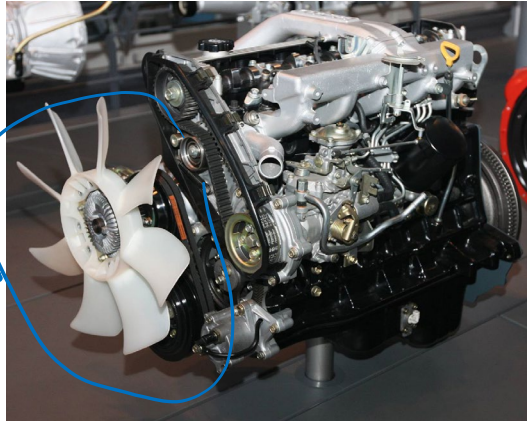
Work input devices: compressors, fans, blowers



<https://medium.com/@johnd/rotary-screw-air-compressor-features-and-advantages-42f82660436c>



<https://www.turbomachinerymag.com/steps-to-reduce-compressor-load-consumption-and-maximize-gas-turbine-flow/>



<https://www.liveabout.com/fan-clutch-going-bad-4173920>

Goal: increase h through work

Work input devices: compressors, fans, blowers

Assume

- steady-flow *
- $\Delta ke = 0$
- $\Delta pe = 0$
- $\dot{Q} = 0$

$$\dot{m}(\Delta h + \cancel{\Delta ke} + \cancel{\Delta pe}) = \cancel{\dot{Q}} - \dot{W}$$

$$\boxed{\dot{m} \Delta h = -\dot{W}}$$


Work done on fluid < 0 $dH < 0, \dot{W} < 0$
 $\Delta h > 0$

Work extraction devices: turbines



<https://www.scientificamerican.com/article/longer-turbine-blades-have-slashed-wind-energy-costs/>



<https://www.turbomachinerymag.com/steps-to-reduce-compressor-load-consumption-and-maximize-gas-turbine-flow/>



<https://energyeducation.ca/encyclopedia/Turbine>

Goal: extract work from fluid.

Work extraction devices: turbines

Assume

- Steady-flow
- $\Delta ke = 0$
- $\Delta pe = 0$
- $\dot{Q} = 0$

$$\dot{m}(\cancel{\Delta h} + \cancel{\Delta ke} + \cancel{\Delta pe}) = \cancel{\dot{Q}} - \dot{W}$$

$$\boxed{\dot{m} \Delta h = -\dot{W}}$$

work done by fluid > 0 , $dH > 0$, $\dot{W} > 0$
 $\Delta h < 0$